

Dhairya Shah Lecture 3 Scribe



L3.pdf
PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** prepared strictly from the provided lecture slides of **CSE400 – Fundamentals of Probability in Computing, Lecture 3: Introduction to Probability Theory**.

No external explanations, interpretations, or examples have been introduced. The content reflects only what appears in the lecture slides, organized for revision purposes as requested.

Lecture Scribe — CSE400: Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Date: January 13, 2026

1. Course Context and Lecture Scope

1.1 Course Identification

- Course Title: **CSE400 – Fundamentals of Probability in Computing**
- Lecture Topic: **Introduction to Probability Theory**
- Instructor: Dr. Dhaval Patel
- Institution: SEAS, Ahmedabad University

This lecture introduces:

- General course information
- Motivation for studying the course
- Engineering applications
- Course structure and participation framework

(As presented in slides 1 and Outline slides)

2. Definitions and Notation (From Lecture Slides)

The lecture slides for this session do **not introduce formal mathematical definitions, notation, probability axioms, or symbolic representations.**

Therefore:

- No definitions of probability, random variables, sample spaces, or events are stated in this lecture.
- No formal notation is introduced in this lecture.

This lecture functions as an introductory and organizational lecture rather than a theoretical one.

3. Assumptions and Conditions

The following assumptions and conditions are explicitly stated or implied through course structure slides:

3.1 Learning Assumptions

- Active participation is expected through Campuswire.
- Learning includes discussion, feedback, and collaborative engagement.
- Students are expected to participate in active learning and class discussions.

(From “Active Learning and Class Discussion” slides)

3.2 Communication Conditions

- Queries should preferably be posted on Campuswire.
- Direct messaging with instructor is allowed for private discussions.
- Contact hours are available through Campuswire.

(From “Timing for Discussion and Difficulty Sessions” slide)

4. Statements of Results or Theoretical Content

This lecture does **not** contain:

- Theorems
- Lemmas
- Propositions
- Mathematical results
- Proof statements

Hence, no formal statements or results are included in this lecture.

5. Step-by-Step Proofs

No proofs are presented in this lecture.

6. Worked Examples (From Lecture Slides)

The lecture includes **contextual examples** rather than mathematical worked examples.

Example 1: Motivation for Learning CSE400

From the outline slide:

- Reason for learning the course:
 - Example: **Daily life conversations**

Step representation (as shown in slides):

1. Probability concepts arise in daily life situations.
2. Conversations involving uncertainty motivate learning probability.

(No mathematical steps are presented in the slides.)

Example 2: Engineering Applications

Engineering applications listed in slides:

- Speech Recognition
- System Radar System
- Communication Network

Step representation (as shown in slides):

1. Probability concepts are applied in engineering domains.
2. These applications motivate study of probability in computing systems.

(No computational or numerical example is shown.)

7. Course Structure Relevant for Examination Preparation

7.1 Active Learning Platform

Campuswire is used for:

- Anonymous participation
- Posting and back-channel communication
- Real-time feedback and polling
- Direct communication with instructor/TAs

(From Campuswire slide)

7.2 Lecture Schedule

Lecture sessions:

- Section 1:
 - 9:30 AM – 11:00 AM (Tuesday, Thursday)
 - GICT Room 136
- Section 2:

- 1:00 PM – 2:30 PM (Tuesday, Thursday)
- GICT Room 137

(From schedule slide)

7.3 Project Component (Overview)

Project weightage:

- 30% of course

Major milestones:

1. Concept Evolution Maps
2. Mathematical Modeling
3. Coding and Simulation
4. Inference and Randomized Algorithms
5. Application of Randomized Algorithms
6. Bounds, Analysis, and Final Deliverables

(From project execution slides)

8. Summary for Revision

From this lecture:

- The lecture introduces course logistics and motivation.
 - No mathematical probability theory content is formally introduced.
 - No definitions, notation, theorems, or proofs are included.
 - Motivation is provided through:
 - Daily life conversations
 - Engineering applications such as speech recognition, radar systems, and communication networks.
 - Active learning and structured participation form part of the course framework.
-

End of Lecture Scribe — Lecture 3

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** prepared strictly from the lecture slides of **CSE400 – Fundamentals of Probability in Computing, Lecture 3:**

Introduction to Probability Theory.

All content is taken only from the provided lecture slides. No external explanations, intuition, examples, or additions have been introduced. The structure follows the requested format for revision purposes.

Lecture Scribe — CSE400: Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Date: January 13, 2026

Prepared strictly from lecture slides for exam revision.

1. Lecture Scope and Context (From Slides)

1.1 Lecture Identification

- Course: CSE400 – Fundamentals of Probability in Computing
- Lecture 3: Introduction to Probability Theory

The lecture content includes:

- General course information
- Motivation for learning the course
- Engineering applications
- Course organization and participation structure

(Title slide and outline slides)

2. Definitions and Notation

2.1 Definitions

The lecture slides for this session do **not introduce formal definitions** related to probability theory.

Specifically, the following are **not defined in this lecture**:

- Probability
- Random variable
- Sample space
- Event
- PMF, PDF, or CDF

Therefore:

Definition Content in this Lecture:

None provided in the slides.

2.2 Notation

The lecture slides do **not introduce mathematical notation** related to probability theory.

3. Assumptions and Conditions (From Slides)

The lecture provides course-related assumptions and operational conditions.

3.1 Learning and Participation Assumptions

From the slides on active learning and Campuswire:

- Participation in discussions is expected.
- Students may participate anonymously.
- Collaborative and active learning is encouraged.
- Real-time feedback through polling is used.
- Direct communication with instructor/TAs is possible.

(Slide: Campuswire – Why use Campuswire?)

3.2 Communication Conditions

From the instructor communication slides:

Step-by-step conditions presented in slides:

1. Contact hours available through Campuswire.
2. Best practice is to put queries on Campuswire.
3. Direct messaging allowed for private conversations.
4. External engagement possible via specified channels.

(Slide: Timing for Discussion and Difficulty Sessions)

3.3 Course Operation Conditions

Lecture sessions:

- Section 1:
 - 9:30 AM – 11:00 AM (Tuesday, Thursday)
 - GICT Room 136
- Section 2:
 - 1:00 PM – 2:30 PM (Tuesday, Thursday)
 - GICT Room 137

(Schedule slide)

4. Statements of Theorems or Results

This lecture contains **no mathematical theorems, propositions, or formal results**.

Hence:

- No theorem statements appear in this lecture.
 - No formal results are introduced.
-

5. Step-by-Step Proofs

The lecture slides contain **no proofs**.

Therefore:

- No proof statements or proof steps are present in this lecture.
-

6. Worked Examples (From Lecture Slides)

The lecture provides conceptual examples only as listed in the outline slides.

6.1 Example: Why Should We Learn CSE400?

From the outline slide:

Example given:

- Daily life conversations

Step-by-step representation as given in slides:

1. Probability concepts arise in daily life contexts.
2. These contexts motivate learning the subject.

(No numerical or mathematical steps are included in the slides.)

6.2 Example: Engineering Applications

Engineering applications listed in slides:

- Speech Recognition
- System Radar System
- Communication Network

Step-by-step representation from slides:

1. Probability is applied in engineering domains.
2. These applications motivate the study of probability in computing.

(No computational or mathematical derivations are provided.)

7. Course Structure Relevant for Examination Preparation

7.1 Active Learning Platform

Campuswire is used for:

- Anonymous participation
- Posting and discussion during lecture
- Collaborative learning
- Real-time feedback
- Direct communication with instructor/TAs

(From Campuswire slide)

7.2 Project Component Overview

Project weightage:

- 30%

Major milestones listed in slides:

1. Concept Evolution Maps
2. Mathematical Modeling
3. Coding and Simulation
4. Inference and Randomized Algorithms
5. Application of Randomized Algorithms
6. Deriving Bounds, Analysis, and Final Submission

(Project execution slides)

8. Revision Summary (Strictly from Slides)

- Lecture 3 introduces course structure and motivation.
- No probability theory definitions, notation, theorems, or proofs are introduced.

- Motivation is provided through:
 - Daily life conversations
 - Engineering applications such as speech recognition, radar systems, and communication networks.
 - Active learning and structured participation form part of the course expectations.
-

End of Lecture Scribe — Lecture 3



I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** prepared strictly from the lecture slides of **CSE400 – Fundamentals of Probability in Computing, Lecture 3:**

Introduction to Probability Theory.

All material is derived only from the provided slides. No external intuition,

interpretation, examples, or additional explanations have been introduced. The structure follows the requested format for revision use.

Lecture Scribe — CSE400: Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Date: January 13, 2026

Prepared strictly from lecture slides for exam revision.

1. Lecture Overview (From Slides)

1.1 Lecture Identification

- Course: CSE400 – Fundamentals of Probability in Computing
- Lecture 3: Introduction to Probability Theory

The lecture introduces:

- General course information
- Learning platform and participation structure
- Motivation for learning CSE400
- Engineering applications
- Course logistics and project overview

(Title slide and outline slides)

2. Definitions and Notation

2.1 Definitions

The lecture slides for this session contain **no formal definitions** related to probability theory.

The following are **not defined** in this lecture:

- Probability
- Random variable
- Sample space
- Event
- PMF, PDF, CDF

Therefore:

Definitions present in lecture: None.

2.2 Notation

The lecture slides do not introduce mathematical notation.

3. Assumptions and Conditions (From Slides)

3.1 Learning Assumptions

From the slides on active learning and Campuswire:

Step-by-step conditions as presented:

1. Participation during lectures is expected.
2. Students may participate anonymously.
3. Learning is collaborative and active.
4. Real-time feedback is provided through polling.
5. Direct communication with instructor/TAs is allowed.

(Slide: Campuswire – Why use Campuswire?)

3.2 Communication Conditions

From instructor interaction slides:

Step-by-step conditions:

1. Contact hours are available through Campuswire.
2. Best practice is to post queries on Campuswire.

3. Direct messaging is allowed for private discussion between student and instructor.
4. External engagement opportunities may be available.

(Slide: Timing for Discussion and Difficulty Sessions)

3.3 Course Schedule Conditions

Lecture sessions listed in slides:

- Section 1:
 - 9:30 AM – 11:00 AM (Tuesday, Thursday)
 - GICT Room 136
- Section 2:
 - 1:00 PM – 2:30 PM (Tuesday, Thursday)
 - GICT Room 137

(Schedule slide)

4. Statements of Theorems or Results

The lecture contains **no mathematical theorems, lemmas, propositions, or formal results**.

Therefore:

- No statements of results are present in this lecture.
-

5. Step-by-Step Proofs

The lecture slides contain **no proofs**.

Hence:

- No proof statements or proof steps are included.
-

6. Worked Examples (From Lecture Slides)

The lecture provides motivation-based examples only.

6.1 Example: Why Should We Learn CSE400?

From outline slide:

Example given:

- Daily life conversations

Step-by-step representation as shown in slides:

1. Probability appears in daily life conversations.
2. This motivates learning the course content.

(No numerical or mathematical intermediate steps are provided in slides.)

6.2 Example: Engineering Applications

Engineering applications listed:

- Speech Recognition
- System Radar System
- Communication Network

Step-by-step representation as shown:

1. Probability concepts are used in engineering applications.
2. These applications motivate studying probability in computing.

(No computational or mathematical steps are provided.)

7. Course Structure Relevant for Revision

7.1 Active Learning Platform

Campuswire is used for:

- Anonymous participation
- Lecture discussion
- Collaborative learning
- Real-time feedback
- Direct communication with instructor/TAs

(From Campuswire slide)

7.2 Project Component Overview

Project weightage:

- 30%

Major milestones listed in slides:

1. Concept Evolution Maps
2. Mathematical Modeling
3. Coding and Simulation
4. Inference and Randomized Algorithms
5. Application of Randomized Algorithms
6. Deriving Bounds, Analysis, and Final Deliverables

(Project execution slides)

8. Revision Summary (Strictly from Slides)

- Lecture 3 introduces course organization and motivation.
 - No probability theory definitions, notation, theorems, or proofs are introduced.
 - Examples are motivational:
 - Daily life conversations
 - Engineering applications such as speech recognition, radar systems, and communication networks.
 - Active learning and structured participation are emphasized.
-

End of Lecture Scribe — Lecture 3



I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems

or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** created strictly from the lecture slides of **CSE400 – Fundamentals of Probability in Computing, Lecture 3:**

Introduction to Probability Theory.

All information is taken only from the provided slides. No external explanations, intuition, examples, or additions have been introduced. The structure follows the requested exam-revision format.

Lecture Scribe — CSE400: Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Date: January 13, 2026

Prepared strictly from lecture slides for exam revision.

1. Lecture Scope and Structure (From Slides)

1.1 Lecture Identification

- Course: CSE400 – Fundamentals of Probability in Computing
- Lecture Topic: Introduction to Probability Theory

The lecture covers:

- General course information
- Course logistics
- Motivation for learning the course
- Engineering applications
- Active learning and participation framework

(Title slide and outline slides)

2. Definitions and Notation

2.1 Definitions

The lecture slides for this lecture do **not contain formal mathematical definitions**.

Specifically, the slides do not define:

- Probability
- Random variable
- Sample space
- Event
- PMF, PDF, or CDF

Therefore:

Definitions included in lecture: None.

2.2 Notation

The lecture slides do not introduce probability notation or mathematical symbols.

3. Assumptions and Conditions (From Slides)

3.1 Learning and Participation Conditions

From the Campuswire and active learning slides:

Step-by-step conditions stated in slides:

1. Participation during lectures is encouraged.
2. Students can participate anonymously.

3. Collaborative and active learning is expected.
4. Feedback on lecture sessions is provided through polling.
5. Personal communication with instructor/TAs is possible through direct messaging.

(Slide: Campuswire – Why use Campuswire?)

3.2 Communication Conditions with Instructor

From discussion and difficulty session slides:

Step-by-step conditions:

1. Contact hours are available 24×7 through Campuswire.
2. Best practice is to post queries on Campuswire.
3. Direct messaging is allowed for private conversations between student and instructor.
4. External engagement opportunities may be available.

(Slide: Timing for the Discussion and Difficulty Sessions)

3.3 Schedule Conditions

Lecture session information:

- Section 1:
 - 9:30 AM to 11:00 AM (Tuesday, Thursday)
 - GICT Room 136
- Section 2:
 - 1:00 PM to 2:30 PM (Tuesday, Thursday)
 - GICT Room 137

(Schedule slide)

4. Statements of Theorems or Results

The lecture slides contain:

- No theorem statements
- No propositions or lemmas

- No mathematical results

Therefore:

Statements of theorems/results in this lecture: None.

5. Step-by-Step Proofs

The lecture slides do not include any proofs.

Hence:

Proofs present in lecture: None.

6. Worked Examples (From Lecture Slides)

The lecture includes motivational examples rather than computational or mathematical examples.

6.1 Example: Why Should We Learn CSE400?

From the outline slide:

Example given:

- Daily life conversations

Step-by-step reasoning as reflected in slides:

1. Probability appears in daily life conversations.
2. This motivates the need to study probability.

(No intermediate mathematical steps are provided in slides.)

6.2 Example: Engineering Applications

Engineering applications listed in slides:

- Speech Recognition
- System Radar System
- Communication Network

Step-by-step reasoning as reflected in slides:

1. Probability concepts are used in engineering applications.
2. These applications provide motivation for learning probability in computing.

(No numerical or algorithmic steps are shown.)

7. Course Structure Relevant for Revision

7.1 Active Learning Platform

Campuswire is used for:

- Anonymous participation
- Lecture discussion
- Collaborative learning
- Real-time feedback
- Direct communication with instructor/TAs

(From Campuswire slide)

7.2 Project Component Overview

Project weightage:

- 30%

Major milestones listed:

1. Concept Evolution Maps
2. Mathematical Modeling
3. Coding and Simulation
4. Inference using randomized algorithms
5. Application of randomized algorithms
6. Bounds derivation, analysis, and final submission

(Project execution slides)

8. Revision Summary (Strictly From Slides)

- Lecture 3 introduces course organization and motivation.
- No probability definitions, notation, theorems, or proofs are introduced.
- Motivation is provided through:

- Daily life conversations
 - Engineering applications such as speech recognition, radar systems, and communication networks.
 - Active learning and structured participation are emphasized as course requirements.
-

End of Lecture Scribe — Lecture 3